

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	I II	Ultrafiltration – glomerulus / Bowman's capsule (1) Reabsorption – proximal convoluted tubule (1)	2			2		
		(ii)		X-{ascending limb/ loop of Henle} and Y-collecting duct (1) Osmoregulation (1)	2			2		
	(b)	(i)		(Conc of urea increases because) {little/no} urea is reabsorbed but water is reabsorbed (1) (Conc of chloride decreases because) Chloride is {reabsorbed /diffuses out /actively transported out}{faster than water) (1)			2	2		2
		(ii)		A. (Concentration of glucose decreases because) glucose is reabsorbed (faster than water) (1) Any three (x1) from B. With inhibitor no respiration and therefore {no/less} ATP (1) C. No active transport of Na ⁺ (out of PCT) (1) D. Therefore no concentration gradient of Na ⁺ / owtte E. For the co-transport of glucose (1)			4	4		
	(c)	(i)		{To compare concentration/ show the loss} (of Na ⁺) {in loop of Henle/ in descending and ascending limbs/ between PCT and DCT/ before and after absorption}			1	1		1